

reduces administrative inefficiencies. Blockchain is also being leveraged by clinical trials and drug supply chains. DApps can also use smart contracts to automate insurance claims and medical billing.

E-governance and voting systems:

DApps are used in decentralised governance and voting systems to help prevent election fraud. Online voting solutions such as Voatz and Horizon State, which are built on blockchain technology, reduce the challenges of traditional voting systems. Because votes are recorded on an immutable blockchain ledger, they cannot be altered or tampered with, making people more trusting of the democratic process.

Blockchain based DApps wield the power to revolutionise how organisations operate. Decentralised autonomous organisations (DAOs) make use of decentralised apps to spur community-led decisions. Members can propose and vote on changes in governance without resorting to centralised authorities.

DApps are likely to become more influential in the future as blockchain technology continues its natural evolution and adoption.

Open source platforms for blockchain development

Here are the key programming platforms for blockchain development, using which a range of secured and decentralised applications can be developed and deployed.

Ethereum

<https://ethereum.org>

Ethereum is an open source blockchain platform and one of the most widely used for developing decentralised apps and smart contracts. While Solidity is its main programming language, it gives developers the ability to write and deploy smart contracts. The decentralised aspect of Ethereum allows for peer-to-peer transactions and distributed apps, which is beneficial for several use cases, from DeFi to NFTs and supply chain solutions.

Hyperledger

<https://www.hyperledger.org/>

Hyperledger is a collaborative, open source project hosted by the Linux Foundation that develops enterprise-grade blockchain frameworks. It offers a suite of customisable tools and an exhaustive set of business blockchain frameworks and libraries for organisations that want to use blockchain technology for supply chain development, financial services and IoT solutions.

Popular projects of Hyperledger are:

- **Hyperledger Fabric:** A permissioned blockchain framework intended for enterprise settings.
- **Hyperledger Sawtooth:** A flexible blockchain architecture designed for building distributed ledgers.
- **Hyperledger Indy:** A blockchain-based framework for the management of identity.

Polkadot

<https://polkadot.com/>

Polkadot is a multichain open source blockchain platform that enables different blockchains to interoperate. It aims to build cross-chain decentralised applications (DApps) and address the scalability problems of conventional blockchains. It connects different parachains (independent blockchains) through a relay chain and enables easy communication between them.

Solana

<https://solana.com/>

Solana is a high-performance open source blockchain designed for decentralised apps and cryptocurrencies. It employs a proprietary, proof-of-history (PoH) consensus which facilitates the execution of tens of thousands of transactions per second on the network. It is used for high-speed, scalable DApps, especially in gaming and finance.

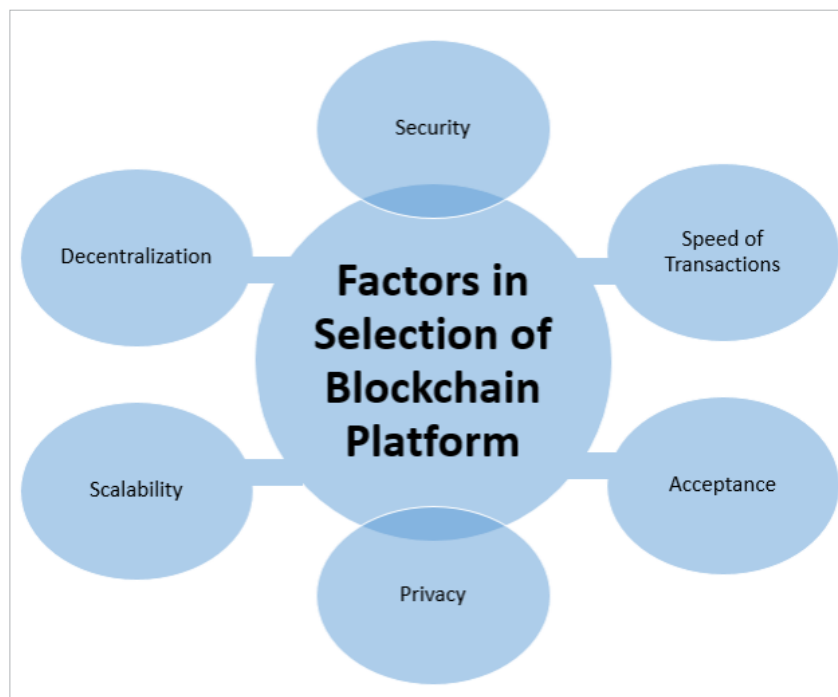


Figure 3: Factors in the selection of a blockchain platform